The Selling closure

Petrus H. Zwart *

Molecular Biophysics and Integrated Bioimaging Division, Lawrence Berkeley National Laboratory, 1 Cyclotron Road, Berkeley, CA 94720, USA.

E-mail: phzwart@lbl.gov

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Abstract

Selling reduction does not select one cell. It returns every obtuse superbase of the lattice — the Selling closure, complete by a theorem of Kurlin (2026) — together with the integer changes of basis between them. This paper shows that the closure is a groupoid and uses that structure for measured cells and for reindexing. The arrows are fixed by the zero pattern of the reduced cell. Each lattice symmetry is one metric-preserving arrow out of each member; only the symmetries that return a member to itself are visible in that reduced cell, and the rest are arrows to other members. The groupoid is generated by Selling flips at vanishing conorms and relabellings, so every lattice symmetry is a word in Selling moves. Its isomorphism types are the 24 Delone sorts, which are derived here from six zero patterns as the closed tie patterns of the closure's slot permutations, with the holohedries as output. In the cubic, hexagonal, rhombohedral and primitive tetragonal and orthorhombic sorts with walls, most of the symmetry maps the reduced cell to another member rather than relabelling it. A measured cell is handled by snapping: its conorms up to 4τ are set to zero, the

exact closure of the resulting lattice is walked, and the members whose conorms in the measured metric exceed $-\tau$ are kept. The result is exactly Selling reduction with its boundary moved to $-\tau$ — unique, equal to the union of the exact closures of all lattices within τ, and complete when the shortest lattice vector exceeds 32τ in squared length — whereas walks on the measured metric miss members for every choice of band. The measured cell's structure is the full subgroupoid of the snapped lattice's groupoid on the members within the band: in-band flips plus bridges through members outside it, with the snapped lattice's symmetries as arrows, each priced in the measured metric. Reindexing reads an arrow of this groupoid from a single reduced cell; under a declared metric error the true operator is proposed, with a computable certificate, and it is lost when the reference is enumerated by snapping only at τ.

1. Introduction

A lattice has infinitely many bases, and reduction is the classical way to choose among them (Niggli, 1928; Delone, 1932). A rule that returns one cell has to decide what to do where two cells are equally good, and there its choice jumps while the lattice changes smoothly. Selling reduction is different in kind. Its result, an obtuse superbase, is unique for a generic lattice, but a lattice with a vanishing conorm — every right angle is one — has several, and Kurlin (2026) proved that the complete set, the Selling closure, has between one and sixteen members according to the Voronoi type and can be enumerated. The closure is therefore the honest output of Selling reduction: the finite set of reduced cells, together with the integer changes of basis that relate them.

This paper makes three points about that output. First, it is a groupoid: its members are the objects and the integer changes of basis between them are the arrows. A lattice symmetry is an arrow that also preserves the metric: it carries each reduced cell either onto itself, as a relabelling of that cell, or onto another reduced cell. The groupoid has a

presentation by Selling moves, and its types are the 24 Delone sorts, which follow from it without tables. Second, a measured cell sits near walls rather than on them, so its exact closure jumps with the noise; snapping the cell onto its nearest stratum recovers the structure exactly, with uniqueness, overlap and completeness statements, and the symmetries of the snapped lattice survive as arrows with a price. Third, reindexing one cell onto another is reading an arrow of this groupoid, and the completeness of the snapped closure is what guarantees that the true operator is among those read.

2. Background

2.1. Selling reduction and the closure

A superbase is a quadruple $F = (v_0, v_1, v_2, v_3)$ of lattice vectors with $v_0 + v_1 + v_2 + v_3 = 0$, any three of which form a basis. Its six conorms are $p_{ij} = -v_i \cdot v_j$, and F is obtuse when all six are non-negative (Selling, 1874; Conway & Sloane, 1992). The metric tensor in the basis (v_1, v_2, v_3) is linear in the conorms, and every lattice vector $x = \sum c_i v_i$ satisfies Selling's identity

$$|x|^2 = \sum_{i<j} p_{ij}\,(c_i - c_j)^2, \tag{1}$$

whatever the signs of the conorms. A *flip* at the slot ij replaces v_i by $-v_i$ and the two remaining vectors v_k, v_l by $v_i + v_k$ and $v_i + v_l$; Selling reduction applies it at negative conorms until none remain, and at a zero conorm it maps one obtuse superbase to another. The Selling closure $\mathcal{G}(\mathbf{G})$ is the set of all obtuse superbases; Kurlin (2026) proved it complete and listed it for the five Voronoi types. Two facts from that work are used throughout (Conway & Sloane, 1992; Kurlin, 2026, Lemmas 2.4, 2.7 and A.1): the seven partial sums of a superbase represent the seven non-zero classes of $\Lambda/2\Lambda$, and a superbase is obtuse exactly when each partial sum is a shortest vector of its class. The lattices sharing a zero pattern form a *stratum*; on a stratum the closure's members are fixed integer superbases.

2.2. Delone sorts

Delone (1932) divided lattices into 24 sorts by the pattern of their reduced Selling parameters — which vanish and which are equal — and Andrews *et al.* (2023) list the 24 patterns. Each sort lies in one Bravais type, and a Bravais type may comprise several sorts.

2.3. Groupoids

A groupoid is a category in which every arrow is invertible (Brandt, 1927; Brown, 2006; Weinstein, 1996). It is connected when any two objects are joined by an arrow, and a connected groupoid is determined, up to isomorphism, by the group of loops at one object and the number of objects: it is generated by the loops at one object together with a spanning tree of arrows (Brown, 2006). The structures used below are of this kind.

3. Methods

3.1. The closure groupoid

The objects are the members of $\mathcal{G}(\mathbf{G})$, each an integer superbase up to the order and common sign of its four vectors. An arrow from member i to member j is an integer change of basis that carries the four vectors of i onto those of j; composition and inverses are those of integer matrices, and the arrows from a member to itself are its relabellings. An arrow is a *symmetry arrow* when it also preserves the metric. A lattice symmetry g — an integer matrix with $g^{\mathsf{T}}\mathbf{G}g = \mathbf{G}$ — maps every obtuse superbase to an obtuse superbase, so it gives exactly one symmetry arrow out of each member i, ending on the member $g(i)$. When $g(i) = i$ the arrow is a loop: g only relabels that reduced cell, and it is the part of the symmetry visible in that cell. When $g(i) \neq i$ the arrow ends on another member, and the symmetry is invisible in

cell i alone. The symmetry arrows out of one member therefore number $|\,\mathrm{Aut}\,|$, and the loops among them form the stabiliser of that member. For each Delone sort the groupoid was computed from a generic lattice of its Selling pattern (Andrews *et al.*, 2023): its members, the automorphisms obtained by testing every member under every relabelling and sign against the metric, and the part of the automorphism group that fixes the reduced superbase.

3.2. Deriving the sorts

The derivation uses no table. For each zero pattern whose complement connects the four vectors (six patterns up to relabelling), the closure of a generic lattice on that stratum is enumerated. A flip at a zero permutes conorm values, so each member under each relabelling acts on the six slots as a permutation; those that keep the zero positions form a finite group Π_Z. A tie pattern — a partition of the non-zero slots into classes of equal conorms — is *closed* when the elements of Π_Z preserving it have exactly its classes as orbits. Closed patterns are collected over all zero patterns and identified when they are the same lattice seen from different members (the smallest pattern string over all members and labellings of the closure).

3.3. Measured cells: snap, walk, filter

The widened closure of a metric at band $\tau \geq 0$ is

$$\mathcal{S}_\tau(\mathbf{G}) = \big\{\, F \;:\; p_{ij}(F;\mathbf{G}) \geq -\tau \text{ for all six slots} \,\big\}, \tag{2}$$

the set of stopping points of Selling reduction with its boundary at $-\tau$. The depth of a superbase is the magnitude of its most negative conorm, so $F \in \mathcal{S}_\tau$ exactly when its depth is at most τ. It is computed in four steps: reduce once to an obtuse superbase F_0; *snap* every conorm of F_0 that is at most 4τ to zero, all at once, giving a lattice Λ^* on a stratum; walk the exact closure $\mathcal{G}(\Lambda^*)$ by flips at its zeros; keep the members

whose conorms in the measured metric are at least $-\tau$. The members of $\mathcal{G}(\Lambda^*)$ with the flips between them form a graph; each flip is labelled by the measured conorm it crosses, which is an integer combination of the snapped conorms, and each member by its depth.

3.4. Reindexing from one reduced cell

A reference R is cached as its widened closure with integer coordinates. A new cell N is reduced once; each cached member under each relabelling, paired with N's reduced superbase, proposes an integer operator $\mathbf{P}$ with $\mathbf{G}_N \approx \mathbf{P}^{\mathsf{T}}\mathbf{G}_R\mathbf{P}$, and its residual is measured in R's frame. A metric error is declared as a Frobenius norm η in R's reduced frame and translated into τ as in Appendix B.

3.5. Verification

Every quantitative statement is checked by code: closure members against all superbases with coordinates in $[-2, 2]$ relative to F_0 (135 408 unimodular bases), filtered by equation (2); groupoid and sort computations by exhaustive enumeration over members, relabellings and signs. Test lattices are given by conorm vectors, so F_0 is obtuse by construction. The exact worked example is the monoclinic cell of Section 4.1; the measured one is the monoclinic cell 30.02, 39.97, 50.03 Å, 90.25°, 92.05°, 89.80°, a 14.4 Å^2 perturbation of the cell 30–40–50 Å, $\beta = 92°$, which sits on two walls. The checks are the test files of Appendix C, cited as [z*nn*].

4. Results

4.1. A primitive monoclinic lattice

Take the monoclinic cell $a = 30$, $b = 40$, $c = 50$ Å, $\beta = 100°$. Its reduced superbase is $M_0 = (-a - b - c,\ a,\ b,\ c)$, with conorms $p_{01} = 639.5$, $p_{02} = 1600.0$, $p_{03} = 2239.5$, $p_{13} = -a \cdot c = 260.5$ and $p_{12} = p_{23} = 0$ Å^2. The two zeros are not near-coincidences:

they are the right angles at the twofold axis b, and every monoclinic P lattice has them. No other conorm is small — the nearest further wall is $260.5\,\text{Å}^2$ away — so the lattice is not close to a higher symmetry, and no frame outside its exact closure enters the widened closure for any band below $260.5\,\text{Å}^2$.[z46]

The closure has six members (Table 1). Each has two zero conorms, and a flip at either leads to a neighbour, so the six form a ring, M_0–M_1–M_5–M_4–M_3–M_2–M_0. The integer changes of basis around the ring are the arrows of the groupoid; any two members are joined by one, and composing them along the ring is consistent.

Table 1. *The Selling closure of the monoclinic cell $a = 30$, $b = 40$, $c = 50$ Å, $\beta = 100°$: its six members, each a superbase up to the order and common sign of its vectors, and the flips at zero conorms that connect them. Every member carries the same six conorm values in a different order.*

member	four vectors	zero-conorm flips lead to
M_0	$-a-b-c,\ a,\ b,\ c$	M_1, M_2
M_1	$-b-c,\ -a,\ b,\ a+c$	M_0, M_5
M_2	$-a-c,\ a+b,\ -b,\ c$	M_0, M_3
M_3	$a+c,\ b-c,\ -b,\ -a$	M_2, M_4
M_4	$a-b+c,\ -c,\ b,\ -a$	M_3, M_5
M_5	$a-b,\ c,\ b,\ -a-c$	M_1, M_4

The lattice has four symmetries, the point group $2/m$, and each is one metric-preserving arrow out of M_0 (Table 2).[z46] The identity and the inversion map M_0 onto itself: they are loops, visible in the reduced cell as relabellings of its four vectors. The twofold axis and the mirror are not. The twofold sends $a \mapsto -a$, $b \mapsto b$, $c \mapsto -c$, and so carries $M_0 = (-a-b-c, a, b, c)$ to $(a-b+c, -a, b, -c)$, which is not a reordering of M_0's vectors up to sign: it is the member M_4, three flips around the ring. In this frame, then, half of the point group cannot be seen in the reduced cell at all; it lives in the arrows between reduced cells. Seen from M_4, the same two symmetries are arrows back to M_0, and the identity and inversion are again the loops.

Table 2. *The symmetries of the monoclinic lattice of Table 1 as arrows out of the reduced cell M_0. Matrices act on the coordinates in the basis a, b, c.*

symmetry	matrix	image of M_0	kind of arrow
1	$\mathrm{diag}(1,1,1)$	M_0	loop (a relabelling of M_0)
$\bar{1}$	$\mathrm{diag}(-1,-1,-1)$	M_0	loop (a relabelling of M_0)
2 along b	$\mathrm{diag}(-1,1,-1)$	M_4	arrow to another member
m normal to b	$\mathrm{diag}(1,-1,1)$	M_4	arrow to another member

4.2. Selling reduction is a groupoid

The closure groupoid is connected, and it is generated by two kinds of move: Selling flips at zero conorms, which connect the members, and relabellings of a member's four vectors with a global sign. Every change of basis between members is a word in these moves, and so is every symmetry, which maps the reduced superbase to some member and is therefore a flip path followed by a relabelling (Appendix A). The symmetry arrows out of any member, one per lattice symmetry, number $|\,\mathrm{Aut}\,|$, and for every one of the 24 sorts this is the order of the holohedry: 48 for the cubic sorts, 24 for hexagonal, 16 for tetragonal, 12 for rhombohedral, 8 for orthorhombic, 4 for monoclinic and 2 for triclinic (Table 3).[z46]

Table 3. *The closure groupoid of each Delone sort, for a generic lattice of its Selling pattern (Andrews* et al.*, 2023; $s_1 \dots s_6$, equal letters equal values). Members are counted up to labelling and sign. "Fix F_0" is the number of symmetries that relabel the reduced superbase itself; the remainder map it to another member. "Reached" is the number of members onto which the symmetries map F_0. Both depend on which member is reduced to; the values are for the frame of the pattern shown.*

sort	Bravais	pattern	zeros	members	| Aut |	fix F_0	reached
C1	cI	rrrrrr	0	1	48	48	1
C3	cF	rr0rr0	2	3	48	16	3
C5	cP	000rrr	3	16	48	12	4
H4	hP	00rrrs	2	6	24	4	6
T1	tI	rrsrrs	0	1	16	16	1
T2	tI	rr0rrs	1	2	16	8	2
T5	tP	000rrs	3	16	16	4	4
R1	hR	rrrsss	0	1	12	12	1
R3	hR	rr0sr0	2	3	12	4	3
O1A	oF	rrsrrt	0	1	8	8	1
O1B	oI	rstrst	0	1	8	8	1
O2	oI	rs0srt	1	2	8	4	2
O3	oI	rs0rs0	2	3	8	8	1
O4	oS	00rsst	2	6	8	4	2
O5	oP	000rst	3	16	8	2	4
M1A	mC	rrsttu	0	1	4	4	1
M1B	mC	rstrsu	0	1	4	4	1
M2A	mC	rs0stu	1	2	4	2	2
M2B	mC	rs0rst	1	2	4	4	1
M3	mC	rs0ts0	2	3	4	4	1
M4	mP	00rstu	2	6	4	2	2
A1	aP	rstuvw	0	1	2	2	1
A2	aP	rs0tuv	1	2	2	2	1
A3	aP	rs0tu0	2	3	2	2	1

The two halves of the groupoid depend on different things. The arrows — which members exist and how they are related — are fixed by the zero pattern, because on a stratum the members are fixed integer superbases; the symmetry arrows by the Bravais type. Several Bravais types share one arrow structure: monoclinic P, orthorhombic C and hexagonal P all have two zeros and six members, distinguished only by their symmetry arrows (4, 8 and 24 out of each member). Conversely, one Bravais type can occupy several arrow structures: body-centred tetragonal lattices fall in T1 or T2, and C-centred monoclinic lattices in five sorts with three different zero patterns, according to the cell's shape. Within a sort the shape does not enter at all: two lattices of one sort have the same members, the same changes of basis and the same symmetry arrows,

and their shape supplies only the metric quantities — conorm values, distances and prices — carried on that structure.

A single reduced cell shows only part of the symmetry. In the sorts with walls, the symmetries that relabel the reduced superbase itself are often a small fraction of the group — 2 of 8 for primitive orthorhombic, 4 of 16 for primitive tetragonal, 12 of 48 for primitive cubic, 4 of 24 for hexagonal, 4 of 12 for rhombohedral R3 — and the rest map it to another member of the closure (Table 3).[z46] These are the operators that reindexing and symmetry assignment need, and they are arrows of the groupoid rather than properties of one cell.

4.3. The 24 Delone sorts from the groupoid

The derivation of Section 3.2, starting from the six zero patterns and nothing else, returns exactly 24 closed tie patterns, and they are the 24 Delone sorts of Andrews *et al.* (2023), with no sort missing and none extra.[z48] The automorphism groups are not supplied; they come out as the holohedries listed above, and the member counts as Kurlin's 1, 2, 3, 6 and 16.[z48] The three patterns that Andrews *et al.* describe as degenerate boundaries (O3, M2B, M3) are closed patterns like the others. In this form a Delone sort is a closed pattern of ties in the closure groupoid of a stratum, counted once across the members of its closure: the zero pattern fixes the arrows, the ties fix the symmetry arrows, and closedness says that a generic lattice with those ties has exactly that symmetry.

4.4. Measured cells: the snap

Uniqueness. Equation (2) mentions no starting cell, reduction order or frame, so $\mathcal{S}_\tau(\mathbf{G})$ is unique. The frame at which reduction to $-\tau$ stops is not: from 400 random unreduced bases of the worked cell at $\tau = 43.2\,\text{Å}^2$ it stopped at six different frames,

and on a noisy orthorhombic cell at sixteen; every stopping frame was a member of $\mathcal{S}_\tau$, but on a noisy cubic cell at $\tau = 10\,\text{Å}^2$ the 400 restarts found 15 of the 16 members.[z45] The object to compute is the set.

Overlap and truncation. For a superbase F let $\delta_F(\mathbf{G}, \mathbf{G}')$ be the largest change of F's own six conorms between two metrics; it involves no frame. Then

$$\mathcal{S}_\tau(\mathbf{G}) = \{\, F \;:\; F \in \mathcal{G}(\mathbf{G}') \text{ for some } \mathbf{G}' \text{ with } \delta_F(\mathbf{G}, \mathbf{G}') \le \tau \,\} \tag{3}$$

(Proposition 1, Appendix A). A member of the widened closure is precisely a frame that is exactly reduced for some lattice consistent with the measurement at band τ. The inclusion $\supseteq$ is the overlap lemma, the inclusion $\subseteq$ the truncation lemma: clipping a member's negative conorms to zero gives the nearest lattice for which it is exactly obtuse.

Completeness. If the shortest lattice vector has squared length $\lambda_1 > 32\tau$, the snap–walk–filter construction returns exactly $\mathcal{S}_\tau(\mathbf{G})$ (Theorem 2, Appendix A). The proof rests on two inequalities: a superbase not obtuse for Λ^* exceeds the class minimum in some class by more than 16τ, and a superbase in the band by at most 16τ, the second through two integer constants on four points, 4 and 8, both attained.[z44] On every test lattice and band, walls placed in the grey zone between τ and 4τ included, the construction returned the brute-force set.[z44] The hypothesis is sufficient, not sharp: the worked cell has $\lambda_1/32 = 28.2\,\text{Å}^2$, and the construction is exact there at 43.2 and $60\,\text{Å}^2$.[z44]

All walls at once. Snapping is simultaneous: the closure of any partial snap is contained in the closure of the full snap, so partial snaps add nothing, and on the worked cell and a noisy cubic cell the union of all eight partial snaps' closures is exactly the full snap's sixteen members.[z46] The threshold 4τ is what the proof requires; on dense corner grids snapping at 2τ is already exact, and snapping at τ is not.[z46]

Walks on the measured metric fail. Flipping wherever a measured conorm is within

a band of zero misses members for every band (Table 4).[z45] The cause is a member reachable only through a non-member: with walls at 1.2τ and 0.5τ, a flip at the first gives a conorm of -1.2τ, outside the band, and a second flip lands on a member of depth 0.7τ. On the idealised lattice both walls are zero and both flips cost nothing. Snapping only at τ misses the same member, returning six of seven on orthorhombic, cubic and plate cells.[z44]

Table 4. *Procedures against brute force on 80 cells with walls at grey-zone and random positions (orthorhombic, cubic, plate, monoclinic and hexagonal strata;* $\tau = 0.1$ *and* $0.5\,\lambda_1/32$*).*

procedure	cells with a wrong set
walk at $\lvert p\rvert \le \tau$, keep only frames within the band	13
walk at $\lvert p\rvert \le \tau$, filter at the end	13
walk at $\lvert p\rvert \le 2\tau$, keep only frames within the band	13
walk at $\lvert p\rvert \le 2\tau$, filter at the end	2
snap at 4τ, walk $\mathcal{G}(\Lambda^*)$, filter	0

Hidden members. The members missed by snapping at τ have a precise location. Their worst conorm is not a wall of F_0 but a difference of two of its small conorms — a near-tie in F_0's frame, which is a wall in their own. They occur only at corners where three walls meet in a triangle (the orthorhombic and cubic corners), exactly when one wall $z \le \tau$ and the other y satisfies $\tau < y \le z + \tau$, and never at the two-wall corners of monoclinic or hexagonal cells.[z46] The oblique wall carries no symmetry: a lattice lying on it has only $\pm\mathbf{I}$.[z46] A hidden member is an alternative reduced frame inside the band, not a pseudo-symmetry — but a second measurement of the same lattice can reduce into it.

4.5. The near closure is a subgroupoid

The structure of a measured cell is the full subgroupoid of Λ^*'s closure groupoid on the members within the band. Its arrows are exact integer changes of basis, taken from Λ^*, and inherit composition and consistency; they are generated by relabellings, the flips whose both ends lie in the band, and one *bridge* for each further component of the

band — a short word of flips through members outside it. Bridges exist exactly where hidden members are: in the corner example, the six in-band members around F_0 form one component and the hidden member another, joined by two flips through a member of depth p_{12}. Its symmetry arrows are those of Λ^*; in the measured metric they are not exact, and each carries a price, the residual $\|\mathbf{P}^{\mathsf{T}}\mathbf{G}\mathbf{P}-\mathbf{G}\|$. The isomorphism type of the near closure is therefore the stratum of Λ^* together with the set of its closure's cells that lie within the band.

One walk gives that set for every band. The worked cell's reduced conorms are $(49.5, 1588.9, 2444.7, 4.2, 847.5, 4.5)\,\text{Å}^2$; snapped at 4×43.2, all three small conorms become walls, Λ^* is orthorhombic with sixteen members, and their depths in the measured metric are 0, 4.2, 4.5, 4.5, 8.7, 8.7, 45.0, 49.5, 49.5, 53.7 (four), 57.9, 62.5 and $62.5\,\text{Å}^2$. The widened closure has six members for $8.7 \le \tau < 45.0$ — the six members of the parent cell's exact closure, at the same integer coordinates — seven at 45.1, nine at 50 and fourteen at $60\,\text{Å}^2$.[z45] The depth at which a frame enters is the band at which the near-symmetry it carries becomes visible: the frames entering at $45.0\,\text{Å}^2$ carry the parent's near-orthorhombic symmetry.

4.6. Reindexing reads an arrow

The operators relating two settings of a lattice are one operator composed with the lattice symmetries, a coset of Aut, and each is read as an arrow between the members the two settings reduce to. On the worked pair at $\tau = 3\eta = 43.2\,\text{Å}^2$ the six-by-six member pairings give twelve operators at $14.4\,\text{Å}^2$ — four distinct operators forming the $2/m$ coset, at a residual equal to the perturbation — and nothing below $1378\,\text{Å}^2$. Once the band passes $45.0\,\text{Å}^2$ the closure admits the frames that propose the pseudo-mmm operators: at $\tau = 60\,\text{Å}^2$, fourteen members give twelve operators at 14.4, ten at 150.7 and nothing below $992\,\text{Å}^2$.[z39] The band decides which near-symmetries are

proposed, and the closure says exactly which.

A new cell needs no closure of its own: its single reduced superbase is paired with the reference's cached widened closure. For fifteen references at orthorhombic, cubic, plate, monoclinic and hexagonal corners, each with twelve new cells carrying a random metric error of declared size and a random reindexing, the new cell's reduced superbase was a member of the reference's widened closure at its true coordinates, and the true operator was among those proposed, in all 180 cases, with the band at the coverage bound of Appendix B.[z47]

Completeness of the cache matters for the coset. A new cell constructed so that its only reduced superbase is the hidden member of an orthorhombic corner is matched from that superbase against two caches. The cache built by Theorem 2 proposes the operators at 7.83, 8.01 and 8.84 Å^2, each with its negative; the cache built by snapping only at τ proposes the first two and loses the third, which is $\pm\mathbf{I}$, the true operator.[z46] The best match is still found, through a near-symmetry, but the coset reported is incomplete.

That the true operator is proposed can be certified before matching. If the new cell's metric, written in the reference frame, differs from the reference by at most η, and V is the summed squared length of the new cell's reduced basis, then the true frame's depth in the reference metric is at most $\sigma = \sqrt{2}\,\eta\,V/(\lambda_{\min}(\mathbf{G}_R) - \eta)$; with $\tau \geq \sigma$ and $\lambda_1 > 32\tau$ the true operator is proposed (Proposition 3, Appendix A). The certificate held in all 180 cases, but it is conservative: its median is 13η, and it lies below the coverage band in 15 of the 180.[z47]

5. Discussion

The mathematics used here is classical. Groupoids go back to Brandt (1927); the closure groupoid is the action groupoid of the integer changes of basis on the Selling cells

that contain a given metric, an object of Voronoi's reduction theory. Crystallography uses groupoids elsewhere, for the partial symmetries of order–disorder structures (*International Tables for Crystallography*, 2006, Vol. C, ch. 9.2), but not, to my knowledge, for reduced cells. What is new is the use: the presentation of the closure by Selling moves, the derivation of the Delone sorts as its closed tie patterns, the near closure of a measured cell as a subgroupoid of a snapped stratum, and pseudo-symmetry as the priced symmetry arrows of that stratum.

The groupoid also explains a difficulty the field has worked around. A reduced cell, Niggli or Selling, carries only the symmetry that relabels it, which in the sorts with walls is a fraction of the point group (Table 3); the remainder are arrows to other reduced cells. Tolerance-based assignment tests one reduced cell's parameters, and distance methods in S^6 (Andrews *et al.*, 2019; Andrews & Bernstein, 2023) reach other cells through boundary maps. The closure keeps all of them with their integer relations, and the snap makes that structure available for measured data with a completeness statement.

What is proved and what is checked should be kept apart. Uniqueness, the overlap characterisation, completeness of snap–walk–filter, the generation of the groupoid, the nesting of snaps and the reindexing certificate are proved in Appendix A, from Kurlin's results and two integer constants established by exhaustive enumeration. The identification of the 24 sorts and the counts of Table 3 are exhaustive finite computations on generic lattices; a version in exact integer arithmetic would make them proofs. Three statements are checked but not proved: that snapping at 2τ suffices, that the true operator is proposed at the sharper coverage band rather than at the certificate, and that the operators accepted within a band are exactly one operator composed with the symmetries of Λ^* priced within it, which needs the coset statement with its frame factor. The first two hold in every case tested.

The frame enters in one place only. Uniqueness, overlap and completeness are statements about τ in conorms, which need no frame. Translating a declared metric error into τ does: conorm shifts are linear in the metric with constants that depend on the frame in which the error is declared, up to $2\sqrt{3}$ on walls in some cuboid frames, and Appendix B gives the exact identity, a frame-free bound and a covariance route.

6. Conclusions

Selling reduction returns a groupoid. Its arrows are fixed by the zero pattern, its symmetry arrows by the Bravais type — one out of each member for every symmetry, a loop only when the symmetry relabels that member — and it is generated by Selling flips at zero conorms and relabellings, and its 24 types are the Delone sorts, derived from six zero patterns with the holohedries as output. A measured cell is snapped onto its nearest stratum at 4τ; the exact closure walked there and filtered at $-\tau$ is exactly Selling reduction with its boundary at $-\tau$, unique, equal to the union of the exact closures of all lattices within the band, and complete when $\lambda_1 > 32\tau$. The measured cell's structure is the subgroupoid of the snapped lattice on the members within the band, with in-band flips, bridges and the snapped lattice's symmetries as priced arrows, and one walk gives it at every band. Reindexing reads an arrow of that structure from one reduced cell, and the completeness of the snapped closure is what keeps the true operator among those read.

Appendix A
Proofs

Write $e_c(F) = |v_c(F)|^2 - \mu_c$ for the excess of a superbase F in class c of $\Lambda/2\Lambda$, where $v_c(F)$ is its partial sum in that class and μ_c the class minimum.

Proposition 1. ($\supseteq$) If F is obtuse for $\mathbf{G}'$ and $\delta_F(\mathbf{G},\mathbf{G}') \le \tau$, every conorm of F under $\mathbf{G}$ is at least $-\tau$. ($\subseteq$) If $F \in \mathcal{S}_\tau(\mathbf{G})$, the metric whose conorms in the frame of F are $\max(p_{ij},0)$ exceeds $\mathbf{G}$ by a positive semidefinite term, so it is positive definite; F is obtuse for it, at δ_F equal to the depth of F. Each negative conorm must move at least to zero, so no nearer metric makes F obtuse. □

Lemma A1 (class excess identity). Let F be any superbase, $w = \sum s_i v_i$ ($s \in \{0,1\}^4$) a partial sum and w' a vector of its class with coordinates $s + 2k$. Then $|w|^2 - |w'|^2 = -4\sum p_{ij}\gamma_{ij}$, with $\gamma_{ij} = \Delta k(\Delta k + d)$, $\Delta k = k_i - k_j$, $d = s_i - s_j$; each γ_{ij} is a non-negative integer, zero exactly when $\Delta k \in \{0, -d\}$. *Proof.* Equation (1), $(d + 2\Delta k)^2 - d^2 = 4\Delta k(\Delta k + d)$, and $t(t + d) \ge 0$ for integers t and $d \in \{-1, 0, 1\}$.

Lemma A2 (class gap). Let Λ^* be obtained from F_0 by setting its conorms $\le \theta$ to zero, be positive definite, and let m^* be the smallest conorm of F_0 above θ. A superbase not obtuse for Λ^* has $e_c \ge 4m^*$ in some class. *Proof.* In the frame of F_0, with w the partial sum of F_0 in class c — a class minimum for $\mathbf{G}$ and for Λ^* — Lemma A1 gives $e_c(F) = 4\sum p^0_{ij}\gamma_{ij}$, a sum of non-negative terms, and the excess for Λ^* is the same sum over the slots above θ. If those terms vanish in all seven classes, every partial sum of F is a class minimum of Λ^* and F is obtuse for it; otherwise some $\gamma_{ij} \ge 1$ multiplies a $p^0_{ij} \ge m^*$.

Lemma A3 (small excess in the band). If $\lambda_1 > 32\tau$ and every conorm of F is at least $-\tau$, then $e_c(F) \le 16\tau$ in every class. *Proof.* Call a slot large if $p_{ij} > 8\tau$, and let N be the negative slots. The large slots connect the four vectors, since otherwise a split crossed only by slots with $p_{ij} \le 8\tau$ has a partial sum of squared length at most $32\tau < \lambda_1$. With w' a class minimum, Lemma A1 in the frame of F gives $0 \le e_c(F) = 4\sum_N |p_{ij}|\gamma_{ij} - 4\sum_{p_{ij}>0} p_{ij}\gamma_{ij}$, so $\sum_{\text{large}} p\gamma \le \tau\sum_N \gamma$. If a large γ were positive, Lemma A4(ii) would give $8\tau\sum_{\text{large}}\gamma < \tau\sum_N\gamma \le 8\tau\sum_{\text{large}}\gamma$; so the large γ vanish, and Lemma A4(i) gives $e_c \le 4\tau \cdot 4$.

Lemma A4 (two constants on four points). Let L be a set of slots connecting the four vectors and N disjoint from L. (i) If $\gamma_{ij} = 0$ on L, then $\sum_N \gamma_{ij} \le 4$. (ii) $\sum_N \gamma_{ij} \le 8 \sum_L \gamma_{ij}$. Both are attained. *Proof.* In (i) $|\Delta k| \le 1$ on L, so with $k_0 = 0$ all $|k_i| \le 3$, and the maximum is a finite enumeration over the 38 connected slot sets, the sets N and the 14 partial sums. In (ii) the ratio is enumerated for $|k_i| \le 15$; beyond, $Q = \sum_L \Delta k^2 > 75$, and $\sum_N \gamma \le \rho Q + \sqrt{|N|\rho Q}$, $\sum_L \gamma \ge Q - \sqrt{|L|Q}$, with $\rho \le 3 + 2\sqrt{2}$ the largest generalised eigenvalue of $\sum_N \Delta k^2$ against $\sum_L \Delta k^2$, keep it below 8 for every pattern. Equality in (ii): $L = \{01, 02, 13\}$, $N = \{03, 12, 23\}$, $s = (0, 0, 1, 1)$.[z44]

Theorem 2. Let F_0 be obtuse for $\mathbf{G}$, $\lambda_1 > 32\tau$, and Λ^* obtained by setting every conorm of F_0 that is at most 4τ to zero. Then Λ^* is positive definite and $\mathcal{S}_\tau(\mathbf{G}) = \{F \in \mathcal{G}(\Lambda^*) : p_{ij}(F; \mathbf{G}) \ge -\tau\}$. *Proof.* If the slots of F_0 above 4τ did not connect the four vectors, some partial sum would have squared length at most $16\tau < \lambda_1$. The inclusion $\supseteq$ is the definition. For $F \in \mathcal{S}_\tau(\mathbf{G})$, Lemma A3 gives $e_c(F) \le 16\tau$ in every class; the smallest surviving conorm exceeds 4τ, so Lemma A2 makes F obtuse for Λ^*. □

Lemma A5 (generation). The closure groupoid is generated by flips at zero conorms and relabellings, and every symmetry is a word in them. *Proof.* The members are reached from F_0 by flips at zero conorms (Kurlin, 2026; reproduced by every enumeration here), so these flips with the relabellings connect all objects. A symmetry maps F_0 to an obtuse superbase, which is a member; as a matrix it is the flip path from F_0 to that member followed by a relabelling. The symmetry arrows form the subgroupoid of the action of Aut on the members: its connected components are the orbits of members under Aut, and the loops at a member are its stabiliser (Brown, 2006).

Lemma A6 (nesting). If $Z' \subseteq Z$ are sets of conorms of F_0 set to zero, every member of the closure for Z' is a member of the closure for Z. *Proof.* The conorms of every

member of a closure are non-negative integer combinations of the free conorms of its stratum (the stratum proposition of Zwart, 2026; checked in [z46]); setting further free conorms to zero keeps them non-negative.

Lemma A7 (sorts as closed tie patterns). On a stratum Z, the symmetries of a lattice act on the slots through Π_Z, and a lattice with tie pattern P has exactly the symmetries in Π_Z that preserve each class of P. *Proof.* A symmetry maps F_0 to a member with the same conorm vector, so it is an element of Π_Z that fixes the values; the members are complete by Kurlin's theorem, so no symmetry is missed. A permutation fixes a vector of values exactly when it preserves each class of equal values. Hence the tie pattern of a generic lattice of a sort is the orbit partition of its symmetry group — a closed pattern — and every closed pattern is the tie pattern of the generic lattices of one sort.

Proposition 3 (reindexing certificate). Let $\mathbf{G}'_N$ be the new cell's metric written in the reference frame, $\|\mathbf{G}'_N - \mathbf{G}_R\|_F \le \eta < \lambda_{\min}(\mathbf{G}_R)$, let F_N be its reduced superbase with basis vectors of summed squared length V, and let $\tau \ge \sqrt{2}\,\eta\,V/(\lambda_{\min}(\mathbf{G}_R) - \eta)$ with $\lambda_1(\mathbf{G}_R) > 32\tau$. Then F_N, at its true coordinates in the reference frame, is a member of $\mathcal{S}_\tau(\mathbf{G}_R)$, the snap–walk–filter construction produces it, and pairing it with the new cell's reduced superbase proposes the true operator. *Proof.* Let u_i be the coordinates of F_N's basis vectors in the reference frame and $\Delta = \mathbf{G}'_N - \mathbf{G}_R$. Then $u_i^{\mathsf{T}}\mathbf{G}'_N u_i \ge (\lambda_{\min}(\mathbf{G}_R) - \eta)|u_i|^2$, so $\sum |u_i|^2 \le V/(\lambda_{\min} - \eta)$. In F_N's own frame the error is $U\Delta U^{\mathsf{T}}$, of Frobenius norm at most $\sum |u_i|^2\,\eta$, and in its own frame every conorm shift is at most $\sqrt{2}$ times that norm (Appendix B). The conorms of F_N under $\mathbf{G}'_N$ are non-negative, so under $\mathbf{G}_R$ they are at least $-\tau$, and Theorem 2 applies. The proposed operator for the pair is the change of basis between the two copies of F_N, which is the true one. □

Appendix B

From a metric error to a band

A conorm is linear in the metric, so a metric error $\Delta\mathbf{G}$ shifts the conorm of a member with coordinate vectors u_i, u_j by exactly $-u_i^{\mathsf{T}}\Delta\mathbf{G}\,u_j$; by Cauchy–Schwarz in the Frobenius product the shift is at most $c_{ij}\eta$, with $c_{ij} = \|\operatorname{sym}(u_i u_j^{\mathsf{T}})\|_F = [\frac{1}{2}(|u_i|^2|u_j|^2+(u_i\cdot u_j)^2)]^{1/2}$, and an error along $\operatorname{sym}(u_i u_j^{\mathsf{T}})$ attains it. Every exact member of a cached closure survives in the widened closure of the measurement exactly when τ is at least the coverage bound $\tau_{\mathrm{cov}} = \max(c_{ij}\eta - p_{ij})$, read off the cache.[z36] Member vectors are not confined to $\{-1, 0, 1\}$; on two and three walls they carry entries up to 3. On walls the largest constant is $\sqrt{2}$ with one zero, $\sqrt{5}$ with two, and with three zeros $2\sqrt{3}$ in six of the sixteen labelled cuboid frames — those whose zeros form a path with v_0 at an interior vertex — and at most $\sqrt{5}$ otherwise; on positive slots it is at most 7.2. In a $2\sqrt{3}$ frame an adversarial error of norm η drives a member to $-3.46\,\eta$, so $\tau = 3\eta$ is not a worst-case guarantee in every frame.[z36] In a member's own frame the member is the standard superbase, whose largest constant is $\sqrt{2}$, so $\tau = \sqrt{2}\,\eta^*$ with η^* the largest error norm over the members' own frames covers every member; it is uniform but loose, 2.8 times the reduced-frame η in the median case and up to 17 times.[z36] With a covariance Σ for the metric, the variance of each member conorm is the quadratic form of Σ on $\operatorname{sym}(u_i u_j^{\mathsf{T}})$, exact in any frame, and the band can be set slot by slot.

Appendix C
Computational checks

Each mark [z*nn*] refers to `manuscript/appendix_z/test_znn_*.py` in the `agentsg` repository.

mark	what is checked
z36	shift identity; constants by zero pattern and frame; adversarial loss at 3η; frame-free bound
z39	worked pair: widened closure by definition; candidate spectrum at $\tau = 43.2$ and 60
z44	constants 4 and 8; class gap and excess; Theorem 2 against brute force; grey zone; worked cell
z45	reduction endpoints and restarts; walks on the measured metric; depth ladder
z46	closure groupoid of the 24 sorts; coset completeness; hidden members; 2τ; nesting
z47	reindexing under a declared error: membership, true operator, certificate
z48	the 24 Delone sorts derived from six zero patterns

References

Andrews, L. C. & Bernstein, H. J. (2023). *Acta Cryst.* A**79**, 485–498.

Andrews, L. C., Bernstein, H. J. & Sauter, N. K. (2019). *Acta Cryst.* A**75**, 593–599.

Andrews, L. C., Bernstein, H. J. & Sauter, N. K. (2023). SELLA – a program for determining Bravais lattice types. arXiv:2303.03122.

Brandt, H. (1927). *Math. Ann.* **96**, 360–366.

Brown, R. (2006). *Topology and Groupoids*. Charleston: BookSurge.

International Tables for Crystallography (2006). Vol. C, *Mathematical, Physical and Chemical Tables*, ch. 9.2. Dordrecht: Springer.

Conway, J. H. & Sloane, N. J. A. (1992). *Proc. R. Soc. London Ser. A*, **436**, 55–68.

Delone, B. N. (1932). *Z. Kristallogr.* **84**, 109–149.

Kurlin, V. (2026). *A complete isometry classification of 3-dimensional lattices.* Revised manuscript, April 2026. https://kurlin.org/projects/lattice-geometry/lattices3Dmaths.pdf.

Niggli, P. (1928). *Krystallographische und strukturtheoretische Grundbegriffe.* Handbuch der Experimentalphysik, Vol. 7, Part 1. Leipzig: Akademische Verlagsgesellschaft.

Selling, E. (1874). *J. Reine Angew. Math.* **77**, 143–229.

Weinstein, A. (1996). *Notices Amer. Math. Soc.* **43**, 744–752.

Zwart, P. H. (2026). The Selling closure: one finite object for lattice symmetry, reindexing and search. In preparation.

Synopsis

Selling reduction returns a finite set of cells with the integer maps between them, and this set is a groupoid in which every lattice symmetry is an arrow from a reduced cell either to itself or to another reduced cell; its 24 types are the Delone sorts. Snapping a measured cell to its nearest stratum gives the same structure under measurement error, and reindexing reads an arrow of it.